Data Academy 2026 ETL-case exercise

Today’s agenda

- Introductions
- Background for the case
- Introduction to the task
- Getting started

Who we are?

Priit Muldma,
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Data Engineer

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Data Engineer

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Who are you?

Case goal

Task: Create an ETL pipeline from open-source data to publishing layer

Goal of the exercise is to understand how a ETL pipeline is done with common tools used in Solita and get hands-on experience specifically **Azure Data Factory (ADF), Azure Blob Storage, Databricks and dbt.**

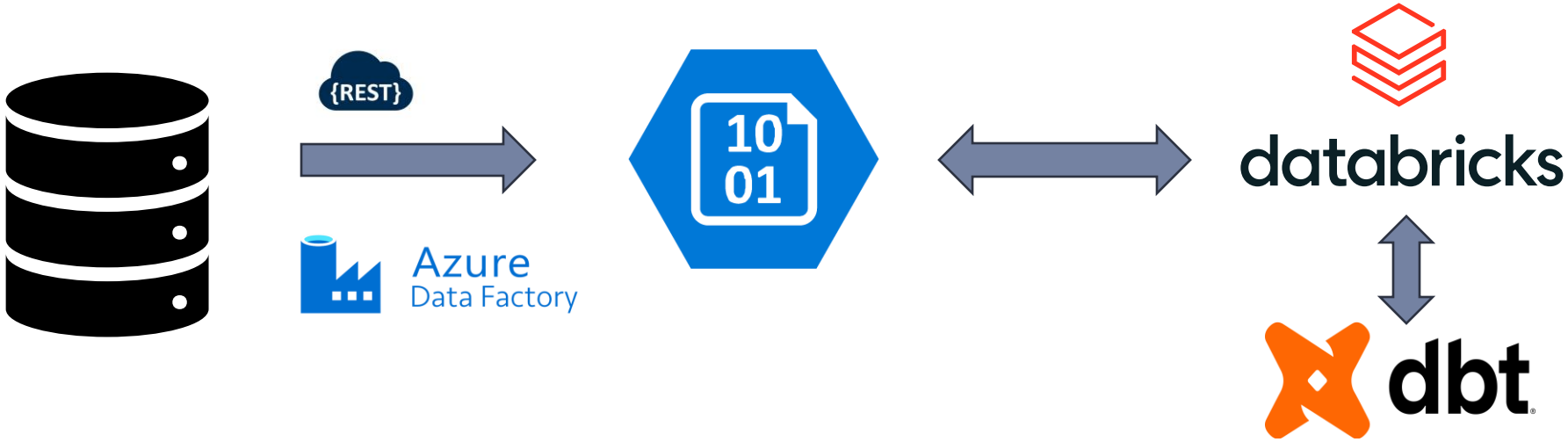
Case participants learn:
- ✅ **Azure resource management:** How to set up and manage resources in Azure.
- ✅ **Get data from the source:** How to fetch data from open API.
- ✅ **Load data into Databricks:** How to load data into cloud.
- ✅ **Transform data:** How to clean and process data for analysis.
- ✅ **ETL process design:** How to build an automated data pipeline.

At the end of this exercise the participants have an Azure-based ETL pipeline that takes FinStat (Tilastokeskus) open data, and transforms and publishes it in analysable form in Databricks.

Case

Analytics department wants to analyze bankruptcies over population by domain(industry), municipality, year and wellbeing services counties.

1. Create Azure Data Factory and Storage Account into Solita sandbox
2. Load data from StatFin open REST API into blob storage using Azure Data Factory
3. Use databricks to "copy" raw source data from CSV-files in the blob storage into Databricks + manual upload the csv file which is coming to slack. We will use Databricks sandbox/catalog: data_academy_fi_06
4. <u>Understand what this data is about!</u>
5. Use dbt to create data models
6. Create a dashboard
7. Schedule daily loads (extra credit)

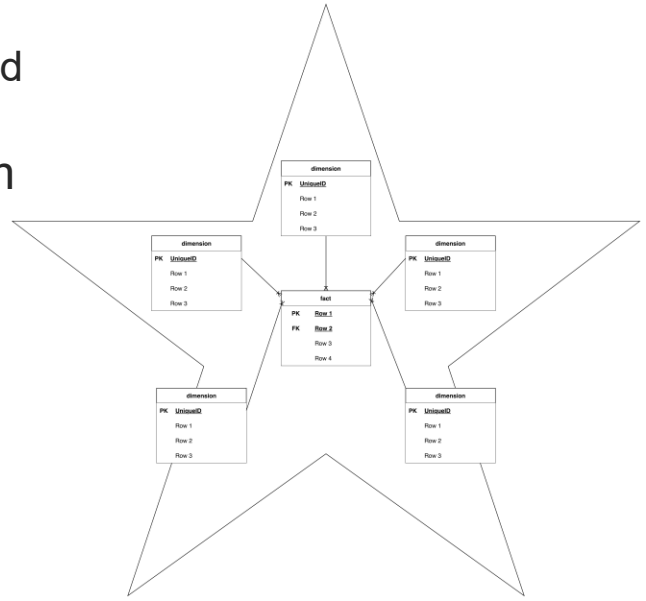


Data sets

- Load three sets of data from StatFin API. Guide will be sent in Slack.
  - **Bankruptcies**
    - Bankrupt enterprises and their employees by domain, municipality and year
  - **Enterprise establishments**
    - Establishment and employee numbers by municipality and year
  - **Population**
    - Population information by municipality and year
- Municipalities and wellbeing services counties as csv-file coming to Slack

Data modelling – dimensional model

- Customer has three business concepts: 1) municipality, 2) domain(industry) and 3) wellbeing services counties
  - Customer wants a cloud warehouse with dimensional data model including dimension and fact tables
- Create entities to publishing layer based on the data from StatFin
  - Entities should have the following numerical attributes by municipality and year:
    - Bankrupt enterprises and laid off employees
      - Number of bankruptcies compared to previous year (both enterprises and employees)
    - Number of enterprise establishments and employees working in them
    - Population



Naming conventions

Vary from project to project, but for clarity in this exercise we will use the following naming conventions

- Azure resources:
  - Resource Group = DataAcademy2026-<yourname>-RG

- Databricks:
  - Schemas= <yourname>_statfin_bronze / <yourname>_statfin_silver etc.
  - Tables = bronze_<sourcesystem>_<tablename> / bronze_<sourcesystem>_<tablename> etc.

- DBT
  - DBT Project = sandbox_<yourname>_databricks

Set up Azure resources

Azure Slides (NOTE! old links)

Azure Hands-on lab You can follow instructions to some degree, as a lot is done with the same way (but not all)

We use Solita Azure Sandboxia

Needed from Azure:

1. Resource Group
2. Azure Data Factory
3. Storage Account + Container (Blob Storage)

(NOTE! Linked Service or datasets are not done to Azure SQL Database (slides 9,10 ja 13,14), because this exercise uses Databricks.

- Databricks Cloud Sandbox login: https://adb-7405608897358843.3.azuredatabricks.net/?o=7405608897358843

- Databricks Best Practises: https://learn.microsoft.com/en-us/azure/databricks/getting-started/best-practices

- Databricks Cloud Hands-on instructions:
- Solita_CaseWork_Hands_on_Instructions.pptx

- Solita Insider Databricks:

  https://insider.solita.fi/sites/tech-partners/SitePageModern/3887/databricks

- DBT Cloud Sandbox login: https://cloud.getdbt.com/enterprise-login/solita-partner/

- DBT Best Practises: https://docs.getdbt.com/best-practices

- DBT Cloud Hands-on instructions: Hands-On Instructions
- Solita Insider DBT: https://insider.solita.fi/sites/tech-partners/SitePageModern/3817/dbt

Help offered

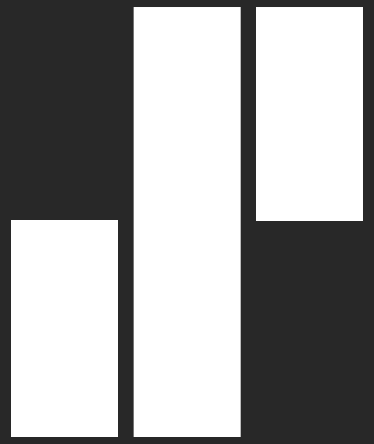
- casework-data-academy-march-2026
- Help or other questions or comments about the case

"Daily" meetings

- Daily (only on selected days) 15min sessions
- Agenda to discuss progress, next steps and blockers
- Calendar invites have been sent

Solution presentations

- Tuesday 2026-04-07
- Round of presentations and following discussions
- Final questions, comments and feedback

Schedule

Questions